

Veerannapet Santhosh Vishal

SDE 2 • Full-Stack • Conversational AI, Agent Platforms & Workflow Engines
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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with nearly 4 years at Kore.ai building AI-native distributed systems — conversational AI agents, agent platforms, durable workflow engines, document intelligence, and microfrontends shipped to 1,000+ enterprise tenants. Strong CS fundamentals from IIIT Hyderabad. Production experience with LLM tool-use, RAG pipelines, MCP integrations, prompt engineering, and ASR/TTS-integrated voice and chat agents. Recognized with Kore.ai **Global Spotlight – January 2026**.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java (OOP, fundamentals), Python, SQL, Bash

Frontend: React, Next.js, Angular, Three.js, RxJS, NgRx, Redux, Monaco, ELK.js, Quill, D3.js, AG-Grid, Tailwind, Module Federation, Canvas API, ReteJS

Backend & APIs: Node.js, Express, Java + Spring Boot (fundamentals), REST APIs, GraphQL, WebSockets, Socket.IO, OAuth2, JWT, Microservices, Event-Driven Architecture

Data & Messaging: MongoDB, PostgreSQL, Redis, Kafka, BullMQ, Restate (durable workflows)

Cloud, DevOps & Observability: AWS, Azure, Docker, Kubernetes, Helm, ArgoCD, Jenkins, Nginx, Linux, Git, Grafana, Prometheus

Testing: Jest, Vitest, Playwright; unit, integration, end-to-end

AI / Conversational: OpenAI, Anthropic, Azure AI, Docling OCR, LLM tool-use, RAG pipelines, MCP integrations, prompt engineering; Kore.ai XO & SmartAssist (voice agents, chat agents, ASR/TTS integration, intent detection, entity extraction, multi-turn dialogue, conversational journey design)

Architecture & Practice: Distributed Systems, System Design (HLD/LLD), OOP, Design Patterns (Factory, Singleton, Observer, Strategy, Facade), SOLID, DRY, YAGNI; AI Dev Tools — Claude Code, Cursor, GitHub Copilot

Methodologies: Agile, Scrum, JIRA, Git, Code Reviews, CI/CD (Jenkins, Docker)

EXPERIENCE

Kore.ai Software India Pvt Ltd — Hyderabad

Jul 2022 – Present

Software Engineer (SDE 2, Grade A2) — AI for Process team. Rapid promotion to SDE 2 for platform delivery and ownership — among the fastest in cohort.

PROJECTS

ABL Platform — Studio, Workflow Engine, Runtime

Jan 2025 – Present

Next.js, TypeScript, Monaco, ELK.js, Restate, Express, MongoDB, Kafka, Kubernetes, OAuth2, JWT

- **Owned Workflow Canvas end-to-end** — Monaco `{{context.}}` IntelliSense, ELK port-constraint auto-layout for multi-handle nodes, searchable Expression Browser, deep-search JSON viewer, live debug panel with WebSocket step-log streaming — powers 15+ node-type AI workflows.
- **Shipped Integration Detail Panel, Auth Profiles, Connection Catalog, and Trigger UI** (cron, webhook, app) — OAuth onboarding across the platform's 50+ third-party connectors with **MCP-based integrations**.
- **Contributed workflow-engine backend internals** (TypeScript on Express) — expression resolver, step-context schema, output-mapping validator, function executor, trigger engine, and reverse-proxy middleware bridging Runtime to Restate.
- **Designed cross-service JWT contract** — service tokens between Runtime and Workflow Engine; covered auth middleware end-to-end with Vitest.
- **Applied OOP design patterns** (Factory, Singleton, Observer, Strategy, Facade) across workflow engine, connector architecture, and runtime services; followed SOLID principles throughout.
- **Drove cross-team coordination across 4 teams** — authored HLDs, LLDs, data-flow audits; ran code reviews across Studio, Runtime, Workflow Engine, and Connectors.

ProcessAI Microfrontend & Visual Flow Designer

Jan 2024 – Present

Angular, Nx, Module Federation, ReteJS, Canvas API, NgRx, WebSockets

- **Led distributed frontend architecture across 4 Angular microfrontends** with Module Federation — 5,800+ files on Nx monorepo; designed cross-app communication, shared state hydration, and independent deployment pipelines shipped to GA at enterprise scale.
- **Built drag-and-drop Visual Flow Designer** with 15+ node categories — LLM, API, Conditionals, Loops, Variables, Human-in-loop, Browser Automation, Document Intelligence.
- **Built LLM nodes and conversational workflow components** — enabling enterprises to design generative-AI voice and chat agents with custom prompt design, journey branching, conditional flows, and human-in-loop handoffs.
- **Shipped Connection Catalog UI and OAuth onboarding flows** with monitored connected-accounts UI and trigger management.

Document Intelligence & Browser Automation

2024 – Present

Angular, Quill, Socket.IO, AG-Grid, Kubernetes, Docker

- **Recognized with Kore.ai Global Spotlight – January 2026** for shaping the Browser Automation UI — translating complex automation workflows into an interface accessible to both technical and non-technical users.
- **Architected document processing pipeline** integrating 4 OCR engines (Docling, Azure, OpenAI, Anthropic) with intelligent provider routing; applied prompt engineering for structured AI outputs across PDF, DOCX, PPTX up to 512 MB.
- **Delivered Browser Automation Studio** with custom Quill Delta-API editor, AI instruction parser, live variable highlighting, undo/redo, and cross-browser stability.
- **Built pod deployment wizard** — resource planning, simulation, variable substitution, auto-scaling 1 → 10 replicas, and full audit history.

WebSDK Performance Optimization

2023 – Present

JavaScript, Webpack, Code Splitting, Lazy Loading

- **Reduced WebSDK bundle 2.3 MB → 920 KB (60%)** via route-level code splitting and lazy loading across 15+ modules; deployed to 1,000+ enterprise tenants — contributing to **Shining Star Award – Q3 2024** (60% optimization + 350+ production fixes).

SmartAssist Contact Center Platform

2023 – 2024

Angular, WebSockets, NgRx, Angular Material; Kore.ai Bot Platform (XO / SmartAssist)

- **Built enterprise contact center serving 10,000+ concurrent agents** across WhatsApp, Telegram, and Microsoft Teams with real-time chat, voice, and email channels.
- **Designed conversational AI journeys** on Kore.ai's bot platform — integrated ASR/TTS pipelines, intent detection, entity extraction, and LLM-powered multi-turn dialogue management for production agent assignment and customer interactions.
- **Tuned routing and queue management** for sub-200ms agent assignment latency at 10K+ concurrent scale; LLM-powered summarization lifted CSAT by 40% and cut post-call documentation time by 40%.

Campaign Dialer, Unified-XO, and Bot Builder

2022 – 2024

Angular, NgRx, D3.js, WebSockets, Microservices, Jenkins, Docker

- **Built voice-bot-driven outbound campaigns supporting 10,000+ concurrent calls** on event-driven microservices with D3.js dashboards (pickup/abandon rates, ROI); integrated LLMs for real-time call summarization, sentiment analysis, and intent recognition.
- **Cut workflow setup 2 hours → 15 minutes (8x)** via modular Bot Builder UI and 25+ reusable Angular Material components standardized across 3 product suites.
- **Reduced deployment time 60% and MTTD 35%** by centralizing logging (Grafana, Prometheus) and hardening Jenkins/Docker CI/CD; mentored junior engineers and authored cross-team design documents.

EDUCATION & RESEARCH

International Institute of Information Technology, Hyderabad (IIIT-H)

2016 – 2021

Dual Degree: B.Tech in Civil Engineering & MS by Research in Building Science

One of India's premier CS research institutes; completed the full CS curriculum.

CS Coursework: Data Structures & Algorithms (Java), Operating Systems, Object-Oriented Programming (Java), Databases & SQL, Computer Networks, Machine Learning, Artificial Intelligence, Software Engineering

Research Scholar — Lab for Spatial Informatics, IIIT-H

2018 – 2021

- Built ML pipeline for water temperature modeling on the Krishna Basin dataset (regression + gradient-boosting in Python).
- Optimized sewage treatment plant capacity (339 MLD) using ML for spatial demand forecasting.
- Automated watershed delineation pipeline in QGIS + Python — replacing manual GIS analysis.
- Built paraphrase detection with Word Movers Distance over word embeddings (NLP, Python).
- Engineered real-time hand-tracking pipeline with OpenCV in Python.
- Implemented UNIX shell in C — process control, I/O redirection, signal handling.

ACHIEVEMENTS

- **Global Spotlight – January 2026** — Kore.ai company-wide recognition for the Browser Automation UI.
- **Shining Star Award – Q3 2024** — 60% performance optimization and 350+ production fixes shipped to enterprise tenants.
- **Rapid Promotion** — Associate Software Engineer → SDE 2; among the fastest in cohort for platform delivery and ownership.
- **JEE Main 2016 – AIR 5460** (top 0.5% of ~1.2M candidates) — secured direct admission to IIIT Hyderabad's flagship dual degree.